Dynamic Customer Segmentation in Retail using Clustering

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PROJECT SUMMARY

Customer segmentation has become an important analytical practice in the retail sector because it enables businesses to group customers according to shared characteristics and behaviours for targeted marketing, improved customer satisfaction, and stronger competitive performance. However, much of the existing literature remains focused on static segmentation approaches, with limited attention to flexible and dynamic methods that can adapt to behavioural changes over time. This study proposes a dynamic customer segmentation framework that combines cascaded Recency, Frequency, and Monetary (RFM) structures with engineered behavioural features and broader customer descriptors to generate richer and more actionable retail profiles. The study aims to develop a self-updating segmentation system capable of capturing behavioural shifts over time with reduced manual intervention. Segment quality will be evaluated using clustering validation metrics, including the silhouette score and Xie-Beni index, to assess cohesion and separation among customer groups. In addition, the project intends to design an interface for visualizing customer segments and tracking their evolution over time, thereby improving the interpretability and practical usability of the framework for decision-making, customer retention, and personalized marketing.

1 INTRODUCTION

Customer segmentation refers to the process of dividing a business's customers into distinct groups based on shared characteristics, behaviours, or needs so that organisations can design more targeted and effective strategies [1]. In the retail sector, segmentation is important because it helps firms tailor marketing activities, improve customer satisfaction and loyalty, and strengthen overall competitiveness [1], [2]. Retail businesses operate in highly dynamic environments where consumer preferences, purchasing behaviour, and market conditions change continuously. As a result, understanding differences among customers is no longer only a descriptive exercise, but a strategic requirement for decision-making and long-term customer relationship management. A large portion of retail segmentation research has relied on unsupervised learning techniques, especially clustering, to identify hidden structures in customer data. Common segmentation approaches include demographic, geographic, behavioural, and firmographic, while clustering methods such as K-means, hierarchical clustering, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), and fuzzy clustering are frequently used to group customers according to similarity [3], [4], [5], [7], [8].

Among feature representation techniques, the RFM model remains one of the most widely used because it summarizes customer transactions into interpretable measures of recency, frequency, and monetary value [3], [4], [5], [8]. These approaches have proven useful in identifying customer groups such as loyal, new, at-risk, and high-value customers, thereby supporting more informed marketing interventions. Despite these contributions, the literature also highlights important limitations in conventional retail segmentation models. Traditional RFM-based approaches are constrained to a small set of variables and often fail to capture richer contextual characteristics such as demographic differences or longer-term behavioural changes [3], [6]. In addition, many existing clustering models rely on static thresholds or one-time data snapshots, making them less suitable for environments where customer behaviour evolves across time [3], [5], [6]. Even where more advanced frameworks have been proposed, full automation and continuous adaptation remain underdeveloped, leaving a gap between academic models and the real-time needs of modern retail decision-making [6]. This study therefore seeks to promote dynamic customer segmentation in the retail sector by moving beyond static, snapshot-based analysis toward a more automated and time-sensitive understanding of customer behaviour. Drawing on the dynamic segmentation ideas of Sobantu and Isafiade [6], as well as broader work on customer representation and clustering in retail [3], [4], [9], [7], [10], the project proposes a framework that combines cascaded RFM features with synthesized demographic and geographic variables to construct richer customer profiles. The study will use transactional retail data, apply preprocessing and feature engineering, divide the data into chronological batches, initialize a baseline segmentation using K-means, and then update customer segments over time through dynamic fuzzy clustering techniques such as the Modified Dynamic Fuzzy C Means (MDFCM) and Crespo Dynamic Fuzzy C Means (CDFCM). Cluster quality will be monitored using validation measures such as Silhouette scores and the Xie-Beni index. By doing so, the study aims to contribute both methodologically and practically. Methodologically, it extends static customer profiling toward a dynamic segmentation process that can better reflect evolving customer behaviour over sequential time cycles. Practically, it seeks to support more actionable business intelligence in retail by producing customer segments that can inform retention strategies, personalized targeting, and adaptive marketing decisions. The study also intends to incorporate an interactive interface for visualizing segment changes over time, which may improve the accessibility and interpretability of the resulting segmentation outputs for retail users.

2 BACKGROUND

2.1 The Shift Toward Data-Driven Personalization in Retail

The growth of digital retail and e-commerce has significantly changed how businesses understand and target customers. Rather than relying on broad, generalized marketing strategies, modern retailers increasingly depend on data-driven segmentation techniques to identify patterns in customer behaviour and support more personalized decision-making [11]. In this context, customer segmentation has become a strategic tool for tailoring marketing campaigns, improving customer satisfaction and loyalty, and enhancing business competitiveness [1], [2]. The shift is central to this study, as customer segmentation is no longer viewed only as the grouping of customers into categories, but as a data-driven process for generating actionable insights in dynamic retail environments. Gomes and Meisen [4] show that customer segmentation in e-commerce has evolved into a structured analytical process involving variable selection, clustering algorithms, and evaluation strategies, with a strong emphasis on personalized customer targeting. Türkmen [5] similarly demonstrates that machine learning methods can be used to process online retail data and generate useful consumer groupings for marketing purposes. Together, these studies suggest that modern retail increasingly depends on analytical segmentation models that can transform raw transaction records into practical business intelligence.

2.2 Traditional and Behavioural Segmentation Models

Customer segmentation is commonly introduced through traditional segmentation categories such as demographic, geographic, behavioural, and psychographic segmentation [1]. These approaches remain useful because they provide a conceptual foundation for grouping customers by who they are, their geographical location, and how they behave. Son et al. [9] shows that geodemographic segmentation can support localized market targeting by combining demographic information with geographic context. This is important because it demonstrates that segmentation can extend beyond purely transactional data and include broader contextual customer characteristics. However, within retail analytics, behavioural segmentation is often regarded as more directly useful because it reflects actual purchasing activity. The most widely used behavioural representation in the reviewed studies is the RFM framework, which summarizes customers according to recency, frequency, and monetary value [3], [4], [5], [9]. RFM remains dominant because it converts transaction histories into interpretable indicators that help distinguish high-value, loyal, dormant, and potentially lost customers. Studies such as Şentürk et al. [3] and Ashraf et al. [8] illustrate that RFM-based clustering can support meaningful customer profiling and marketing strategy development, especially when paired with methods such as K-means.

2.3 Customer Representation and Clustering in Retail Segmentation

A key theme in the literature is that segmentation quality depends heavily on how customers are represented before clustering is performed. Customer representation is the construction and organization of variables that capture meaningful behavioural patterns and improve both clustering quality and interpretability [4], [9]. Within the reviewed studies, several feature representation strategies appear. RFM remains the most common, but other approaches have also emerged to address data complexity and improve interpretability. For example, Ufeli et al. [7] use Factor Analysis of Mixed Data (FAMD) to handle both numerical and categorical variables, while Rungruang et al. [10] apply Formal Concept Analysis (FCA) to produce a more interpretable hierarchical RFM-based segmentation model. Son et al. [9] further show how Principal Component Analysis (PCA) can be used to reduce redundancy and create a more compact feature space before clustering. In clustering, K-means is the most frequently used segmentation method across recent retail studies because it is simple, computationally efficient, and effective for partitioning customers into distinct groups [3], [5], [8], [7]. Hierarchical clustering is often used for comparison or for revealing layered segment structures [5], [7], [10], while multi-method comparisons are used to assess segmentation robustness across datasets [5], [7], [9].

2.4 Limitations of Conventional Static Segmentation Models

Although conventional retail segmentation approaches remain useful, the literature also identifies major limitations. One of the most consistent criticisms is that the standard RFM model is constrained to only three variables, which may oversimplify customer behaviour and fail to capture richer characteristics such as demographic context or longer-term purchase history [3], [6]. The standard RFM does not naturally reveal segments shaped by important contextual differences among customers. A second limitation is that many traditional models are static. Snapshot-based clustering approaches, including many K-means implementations, are typically built on one-time data extracts and rely on manually defined thresholds or fixed assumptions that do not update well as customer behaviour changes over time [3], [4], [5]. Şentürk et al. [3], Türkmen [5], Ufeli et al. [7], Ashraf et al. [8], Rungruang et al. [10], and Cordova [12] all contribute useful segmentation insights, yet their models remain fundamentally static and do not offer a fully adaptive framework for continuous behavioural updates. Even Cordova's [12] big-data retail study, while rich in descriptive analytics and transactional pattern discovery, still does not implement a true real-time or continuously updating segmentation process. A third limitation is the lack of automation in dynamic customer segmentation. Existing approaches still require considerable manual intervention when constructing or updating segmentation models, particularly when customer behaviour shifts gradually or when new behavioural patterns emerge suddenly. This creates a gap between traditional clustering practice and the real needs of modern retail systems, where customer segments must respond more flexibly to continuous transactional change.

2.5 Advanced Dynamic Clustering and Cascaded Approaches

To address these limitations, more recent studies have begun exploring dynamic and hybrid segmentation frameworks. Sobantu and Isafiade [6], propose a cascaded RFM-based fuzzy clustering model for dynamic customer segmentation in the retail sector. This study is particularly important because it moves beyond conventional static segmentation by integrating cascaded RFM structures with Modified Dynamic Fuzzy C-Means (MDFCM) and Crespo's Dynamic Fuzzy C-Means (CDFCM) [13]. Unlike hard clustering, fuzzy clustering allows a customer to belong to multiple clusters with varying degrees of membership, making it more suitable for retail environments where customer behaviour may overlap or evolve over time. The cascaded RFM idea is also important because it attempts to overcome the rigidity of traditional RFM by decomposing the model into structural pairs such as Recency-Frequency, Recency-Monetary, and Frequency-Monetary. These behavioural outputs are then combined with synthesized demographic and geographic variables to form richer customer profiles. This reflects a broader movement in the literature toward multi-dimensional customer representation rather than flat numeric scoring alone [4], [6], [7], [9], [10]. The dynamic component is then introduced through time slicing, where an initial K-means model is built at Time Cycle 0 and later batches are updated through MDFCM and CDFCM, allowing cluster migration, creation, or removal as behaviour changes.

2.6 Research Gap and Positioning of the Present Study

Overall, the literature shows that customer segmentation in retail has progressed from traditional descriptive grouping toward more advanced clustering-based behavioural analysis. However, the reviewed studies also indicate that most retail segmentation models still rely on static structures, one-time clustering procedures, and limited customer descriptors. While some studies extend segmentation with mixed data methods, geodemographic modelling, or improved interpretability, relatively few provide an adaptive framework capable of continuously updating customer groups as new transactions arrive [3], [12]. This project is positioned within that gap. The proposed system combines cascaded RFM, synthesized demographic and geographic variables, initial K-means clustering, and dynamic fuzzy clustering techniques to create a self-updating segmentation process. In doing so, it seeks to contribute not only an adaptive segmentation model, but also an interface for visualizing customer groups and their changes over time, thereby improving both practical usability and interpretation for retail decision-making. Table 1 presents a summary of research trends in customer segmentation.

Table 1: Summary of customer segmentation in the retail sector

Ref	Main Focus/Topic	Dataset Used	Methodology/Model Used	Key Findings / Contributions	Limitations / Gaps Identified
H. Şentürk et al. (2024) [3]	Retail customer profiling using clustering methods	Textile retail e-commerce dataset with 38,975 customers over one year	RFM scoring with K-means, AGNES, and DBSCAN	Shows that conventional clustering can produce practical retail groups such as loyal, potential, new, and lost customers	Static and snapshot-based; does not continuously adapt to changing customer behavior
M. A. Gomes and T. Meisen (2023) [4]	Review of customer segmentation methods for personalized targeting in e-commerce	Literature review of 105 publications (2000–2022)	Structured review of segmentation process, variables, algorithms, and evaluation methods	Strong review anchor; shows that the literature is dominated by RFM-based representation, clustering, and manual feature handling	Review paper only; does not propose or test a dynamic segmentation model
B. Türkmen (2022) [5]	Machine learning for customer segmentation in online retail	Online retail data	K-means, Hierarchical Clustering, DBSCAN, and traditional RFM grouping	Gives a comparative perspective on unsupervised customer segmentation methods and their business usefulness in retail	Mainly comparative and static; does not address continuous model updating

Ref	Main Focus/Topic	Dataset Used	Methodology/Model Used	Key Findings / Contributions	Limitations / Gaps Identified
S. Sobantu and O. E. Isafiade (2025) [6]	Dynamic customer segmentation for automating continuous transactional updates	South African pharmacy retail transactional dataset (55,150 records; 7 variables)	Cascaded RFM, MDFCM, and CDFCM	Proposes a dynamic cascaded RFM-based segmentation framework and shows strong performance as clusters evolve	Still depends on baseline training data and some user-defined setup, so full automation is not yet achieved
C. P. Ufeli et al. (2025) [7]	Customer segmentation with mixed-type variables using dimensionality reduction	Retail dataset with 3,900 customers	FAMD, K-means, and Agglomerative Clustering	Shows that combining numerical and categorical variables can improve segmentation quality and create richer customer profiles	Still a static clustering framework; does not support real-time behavioral updating
A. Ashraf et al. (2025) [8]	Framework for customer segmentation to improve marketing strategies	Online retail dataset	RFM, K-means, Mini-Batch K-means, Spectral Clustering, and Fuzzy K-means	Strong framework-based study linking segmentation directly to marketing strategy and decision support	Remains centered on conventional static clustering over RFM variables
N. N. Son et al. (2025) [9]	Geodemographic market targeting in major localized cities	Vietnam demographic dataset covering 11,165 administrative units	PCA, K-means, geodemographic modelling	Important for showing that segmentation can go beyond basic RFM and include demographic and geographic context	Focuses more on census and geographic data than transactional customer behavior; not dynamic
C. Rungruang et al. (2024) [10]	Interpretable RFM-based customer segmentation	Online Retail II dataset (UCI Machine Learning Repository)	RFM combined with Formal Concept Analysis (FCA)	Useful for feature engineering and interpretability; shows how segmentation knowledge can be structured more clearly	Still limited by the traditional RFM structure and not fully dynamic
R. Sharma et al. (2023) [11]	Broad review of e-commerce and digital transformation, covering trends, consumer behaviour, emerging technologies, challenges, and business implications.	Secondary data from academic journals, government reports, industry publications, and economic analyses, framed around the Indian economy.	Mixed-method study combining qualitative and quantitative secondary-data analysis; also discusses industry case studies and future research directions.	Shows that digital transformation can improve operational efficiency, market reach, and customer experience; highlights m-commerce, social commerce, voice commerce, AI, blockchain, and IoT as major drivers.	Broad conceptual/review-style paper rather than an empirically tested segmentation study; no primary transactional dataset or cluster-validation experiment is reported, and the scope is partly India-centered. The last two points are inferred from the paper's stated method and scope.
R. S. Cordova (2024) [12]	Big-data-driven customer segmentation in online retail.	Online Retail Ecommerce Dataset; UK-based retailer transaction data	Descriptive analytics, data cleaning, feature engineering, exploratory data analysis, descriptive statistics, and K-means clustering in Python.	Shows that online retail transaction data can support practical customer segmentation; identifies country, product, hourly, daily, and monthly sales patterns, and finds that some clusters have higher sales frequency and sales value.	Claims dynamic, real-time, and demographic-rich segmentation, but the implemented study is mostly static and transaction-based; no true streaming or online update mechanism is shown.
M. Sivaguru and M. Punniyamoorthy (2020) [13]	Dynamic customer segmentation using a modified dynamic fuzzy c-means (MDFCM) clustering algorithm.	Retail supermarket dataset with eleven new data updates.	Modifies Crespo's dynamic fuzzy c-means approach; compares MDFCM against CDFCM and FDFCM; validates using Xie-Beni, WSSE, RMSE, Kwon, and Tang indices, with statistical testing and a case-study DCS framework.	Finds that MDFCM is the most effective algorithm among the compared methods for DCS and demonstrates its usefulness through a proposed dynamic customer segmentation framework.	Evaluation is reported on one retail supermarket dataset with eleven updates; the emphasis is mainly on clustering-quality metrics, so broader managerial interpretability and cross-domain validation remain limited. The second point is an inference from the reported evaluation scope.
This study	Customer segmentation in online retail sector	Online Retail Ecommerce Dataset; UK-based retailer transaction data	Cascaded RFM, K means, MDFCM, and CDFCM	Study still in progress	

3 METHODOLOGY

This section explains the end-to-end methodology used to build the dynamic customer segmentation framework. The framework combines data collection, data exploration, preprocessing, feature engineering, baseline clustering, dynamic updating, post-hoc analytics and dashboard design. The method follows a dual-dataset approach so that the system can test both a business-to-consumer cosmetics context and a business-to-business online retail context. Throughout this section, i denotes the customer index and t denotes the monthly time cycle.

3.1 Data collection and preprocessing

This study uses two retail datasets. Dataset A represents a simulated cosmetics retail business and supports the business-to-consumer branch of the framework. It contains raw transaction records for customers buying cosmetics products. The transaction variables include customer identifier, invoice or transaction identifier, product description, transaction date, quantity and unit price. This dataset is also enriched with generated demographic, geographic and psychographic profile variables such as age, gender, region, climate zone and customer persona so that the framework can test richer customer profiling.

Dataset B represents the UK Online Retail dataset used in the business-to-business branch of the framework. It contains online retail transaction records from a UK-based retailer, including invoice number, stock code, product description, quantity, invoice date, unit price, customer identifier and country. Unlike Dataset A, Dataset B does not include demographic variables; therefore, its segmentation focuses mainly on firmographic and behavioural transaction patterns. Table 2 summarises the datasets and the main features used in the study. The dual data exploration process is illustrated in Figure 1.

Table 2: Summary of datasets used in the dynamic customer segmentation framework.

Dataset	Retail context	Main variables and features
Dataset A	SA Cosmetics retail, B2C	Customer ID, transaction date, product, quantity, unit price, RFM variables, cascaded RFM features, generated demographic, geographic and psychographic attributes.
Dataset B	UK WholeSalers Online Retail, B2B	Invoice number, stock code, product description, quantity, invoice date, unit price, customer ID, country, RFM variables and cascaded RFM features.

Data exploration starts by checking the structure, variable types, missing values, duplicate records, invalid quantities, returns and cancellations. The cleaning stage removes cancelled transactions, returns, missing customer identifiers and duplicate entries. It also standardises date fields and converts transaction values into consistent monetary values. After cleaning, the data are aggregated at customer level and transformed into recency (R), frequency (F) and monetary (M) measures. The framework also creates the cascaded interaction features RF, RM and FM to capture relationships between the original RFM dimensions. Log transformation and standardisation are applied where needed to reduce skewness and place features on comparable scales.

The cleaned customer-level features are stored in an SQLite database. The database keeps a customer profile table, preprocessing results, cycle 0 baseline results, dynamic loop results, migration records and model evaluation outputs. Time slicing is then applied by splitting transactions into chronological monthly batches $t = 0, 1, \ldots, n$. Cycle 0 acts as the initial baseline period, while cycles $t \geq 1$ provide sequential updates for the dynamic clustering loop. This time-sliced structure allows the system to compare customer states across months and detect segment movement.

3.2 System architecture and workflow

The system architecture uses three layers: data collection and storage, data processing and logic, and end-user engagement. The bottom layer stores raw datasets and processed customer states in SQLite. The middle layer performs synthetic data generation, data processing, cascaded RFM feature construction, baseline K-means clustering, dynamic fuzzy clustering and post-hoc Random Forest analytics. The top layer provides an interactive dashboard for the marketing analyst. Figure 2 shows the updated system architecture from the slide deck.

The workflow starts when raw data from Dataset A and Dataset B enter the data processor. The processor cleans the records, creates RFM and cascaded RFM features, and sends the cycle 0 state to the K-means engine. After the baseline model creates the initial clusters, the dynamic continuous loop updates the clusters across later time cycles using MD-FCM and CDFCM logic. The post-hoc analytics component then trains a Random Forest model to identify the drivers of segment membership. The dashboard retrieves customer profiles, segment labels, migration flags and evaluation scores from the database.

3.3 Database design

The SQLite database organises the segmentation pipeline into linked tables. The `customer` table stores unique customer identifiers and profile attributes. The `preprocessing_results` table stores the cleaned RFM, cascaded RFM and scaled feature values. The `cycle0_segmentation` table stores the baseline K-means labels, centroids and WCSS values. The `dynamic_loop_results` table stores updated centroids, membership strengths, final segment labels and migration flags for every cycle. This design keeps the raw, processed and modelled outputs separate while preserving a full customer history across time cycles.

3.4 Data synthesis and feature engineering

Dataset A uses additional synthetic profile variables to make the cosmetics retail scenario more realistic. The framework generates demographic attributes such as age and gender, geographic attributes such as city, province, region and climate zone, and psychographic persona labels such as trend-focused buyer or premium loyal buyer. These features are merged with each unique customer ID. The climate and persona variables support richer segment interpretation, for example identifying climate-sensitive skincare buyers or premium loyal customers.

Dataset B retains only behavioural and firmographic transaction features because the original data do not contain demographic information. This approach avoids adding unsupported personal attributes to a B2B dataset. The full customer state is represented as

$$S_{i,t} = B_{i,t} \oplus P_i, \tag{1}$$

where $B_{i,t}$ contains behavioural features at cycle t, P_i contains stable profile variables where available, and $\oplus$ denotes concatenation. For clustering, the framework extracts a training state $S_{i,t}^{\text{train}}$ that excludes names, contact details and other identifiers. Dataset A uses RFM, cascaded RFM and encoded profile variables; Dataset B uses only behavioural RFM and cascaded RFM variables.

3.5 Initial baseline segmentation with K-means

The first modelling step creates the baseline segmentation at cycle 0 using K-means. The appropriate number of clusters K is selected by testing candidate values from 2 to 10 and calculating the Within-Cluster Sum of Squares (WCSS):

$$\text{WCSS}(K) = \sum_{k=1}^{K} \sum_{S_{i,0}^{\text{train}} \in C_k} \|S_{i,0}^{\text{train}} - V_k\|^2, \tag{2}$$

where V_k denotes the centroid of cluster k. Plotting WCSS against K produces an elbow curve, and the elbow point K^* is selected as the baseline number of clusters. K-means then assigns each customer to the nearest centroid and updates centroids by averaging assigned points. The model minimises

$$J = \sum_{k=1}^{K^*} \sum_{S_{i,0}^{\text{train}} \in C_k} \|S_{i,0}^{\text{train}} - V_k\|^2. \tag{3}$$

The final cycle 0 labels and centroid coordinates are saved to the database and become the reference state for dynamic updates.

Table 3: Pseudocode for the baseline K-means segmentation step.

Algorithm 1: Initial baseline K-means segmentation

Input: Cycle 0 training states $S_{i,0}^{\text{train}}$, candidate values $K = \{2, \ldots, K_{\max}\}$.

Output: Optimal cluster number K^*, baseline labels and centroids.

1. Run K-means for each candidate K.
2. Calculate WCSS(K) using Eq. (2).
3. Select K^* at the elbow point.
4. Train final K-means using K^*.
5. Save baseline labels, centroids and WCSS values to the database.

3.6 Dynamic segmentation loop

The dynamic customer segmentation loop updates customer segments across monthly cycles $t = 1, \ldots, n$. At each cycle, the data processor loads the new monthly transactions, recalculates R, F, M, RF, RM and FM, and constructs the updated training state $S^{\text{train}}_{i,t}$. Dynamic clustering algorithms update clusters as new data arrive, allowing customers to migrate between segments and allowing the system to detect changes in segment structure.

The framework applies MDFCM and CDFCM logic to the updated customer states. In fuzzy clustering, each customer has a membership strength $Se_{i,k,t} \in [0, 1]$ indicating the degree of belonging to cluster k at cycle t. The fuzzy clustering objective function is

$$J_t = \sum_{i=1}^{n_t} \sum_{k=1}^{c_t} (Se_{i,k,t})^m \|S^{\text{train}}_{i,t} - V_{k,t}\|^2, \tag{4}$$

where $m > 1$ controls cluster softness. Membership values are updated using

$$Se_{i,k,t} = \left(\sum_{j=1}^{c_t} \left[\frac{\|S^{\text{train}}_{i,t} - V_{k,t}\|}{\|S^{\text{train}}_{i,t} - V_{j,t}\|} \right]^{\frac{2}{m-1}} \right)^{-1}, \tag{5}$$

and centroids are updated as

$$V_{k,t} = \frac{\sum_{i=1}^{n_t} (Se_{i,k,t})^m S^{\text{train}}_{i,t}}{\sum_{i=1}^{n_t} (Se_{i,k,t})^m}. \tag{6}$$

The sequence diagram in Figure 3 shows this monthly loop, including cycle 0 initialisation, cycles 1 to n dynamic updating and post-hoc interpretation.

Table 4: Pseudocode for the dynamic segmentation loop.

Algorithm 2: Dynamic segmentation loop

Input: Baseline centroids, baseline labels, time cycles $t = 1, \ldots, n$, updated customer states.

Output: Updated segment labels, membership scores and migration records.

1. Load monthly transaction data for cycle t.
2. Recalculate RFM and cascaded RFM features.
3. Construct $S^{\text{train}}_{i,t}$.
4. Apply MDFCM/CDFCM dynamic fuzzy clustering.
5. Update centroids and membership scores.
6. Assign the final label using the highest membership value.
7. Save updated labels, memberships and migration flags to SQLite.

3.7 Segment resolution and post-hoc analytics

After dynamic clustering, numerical cluster identifiers are translated into business-friendly segment labels based on RFM patterns, membership strength and migration behaviour. For example, customers with recent purchases, high frequency and high monetary value are labelled as *High-Value Loyal*; customers with older purchase activity and moderate spend are labelled as *At-Risk*; and customers with low frequency and low monetary value are labelled as *Dormant*. Dataset A uses climate and persona attributes to refine these labels, while Dataset B uses transaction intensity and wholesale purchase behaviour.

The final saved state becomes

$$S^{\text{final}}_{i,t} = S_{i,t} \oplus [C_{i,t}, Se_{i,k,t}, \text{Migrate}_{i,t}, L_{i,t}], \tag{7}$$

where $C_{i,t}$ is the cluster label and $L_{i,t}$ is the business-readable segment label. A Random Forest classifier is then trained post-hoc on final labels to identify the most influential variables driving segment membership. Feature importance is calculated using the average decrease in node impurity:

$$FI_j = \frac{1}{T} \sum_{r=1}^{T} \text{ImpurityDecrease}_{j,r}, \tag{8}$$

where T is the number of trees and FI_j is the importance of feature j.

3.8 Evaluation and validation

The evaluation stage measures both baseline and dynamic segmentation quality. During K-means selection, the elbow method compares WCSS values across candidate K values. After baseline training, the silhouette score assesses cluster cohesion and separation:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \tag{9}$$

where $a(i)$ is the average distance between customer i and customers in the same cluster, and $b(i)$ is the minimum average distance to customers in the nearest different cluster. Scores close to 1 indicate well-separated clusters, scores near 0 indicate overlapping clusters, and negative scores indicate possible misassignment.

During dynamic clustering, the Xie–Beni index measures fuzzy cluster compactness and separation:

$$XB_t = \frac{\sum_{i=1}^{n_t} \sum_{k=1}^{c_t} (Se_{i,k,t})^m \|S^{\text{train}}_{i,t} - V_{k,t}\|^2}{n_t \, \min_{p \neq q} \|V_{p,t} - V_{q,t}\|^2}. \tag{10}$$

Smaller XB_t values indicate more compact and better-separated fuzzy clusters. Migration rate, segment size change and feature importance scores are also tracked so that the dynamic output remains interpretable.

3.9 Interface and dashboard design

The final stage connects the analytics backend to an interactive dashboard. After each cycle, the API queries the SQLite database and retrieves customer profiles, segment labels, migration records, membership scores, WCSS values, silhouette scores, Xie–Beni values and Random Forest feature importance scores. The dashboard displays summary cards such as total customers, number of segments, average spend and churn risk. It also presents charts for segment distribution, average spend, customer migration and key insights. Figure 4 shows the dashboard wireframe.

Marketing analysts use the interface to identify high-value customers, detect at-risk customers, compare segment behaviour across time cycles and design targeted retention strategies. Dataset A supports richer demographic, geographic and psychographic exploration, while Dataset B focuses on firmographic and transactional purchasing behaviour. This interface completes the closed loop by turning the clustering output into accessible decision support.

4 RESULTS AND DISCUSSION

5 CONCLUSIONS

6 ACKNOWLEDGMENTS

APPENDIX

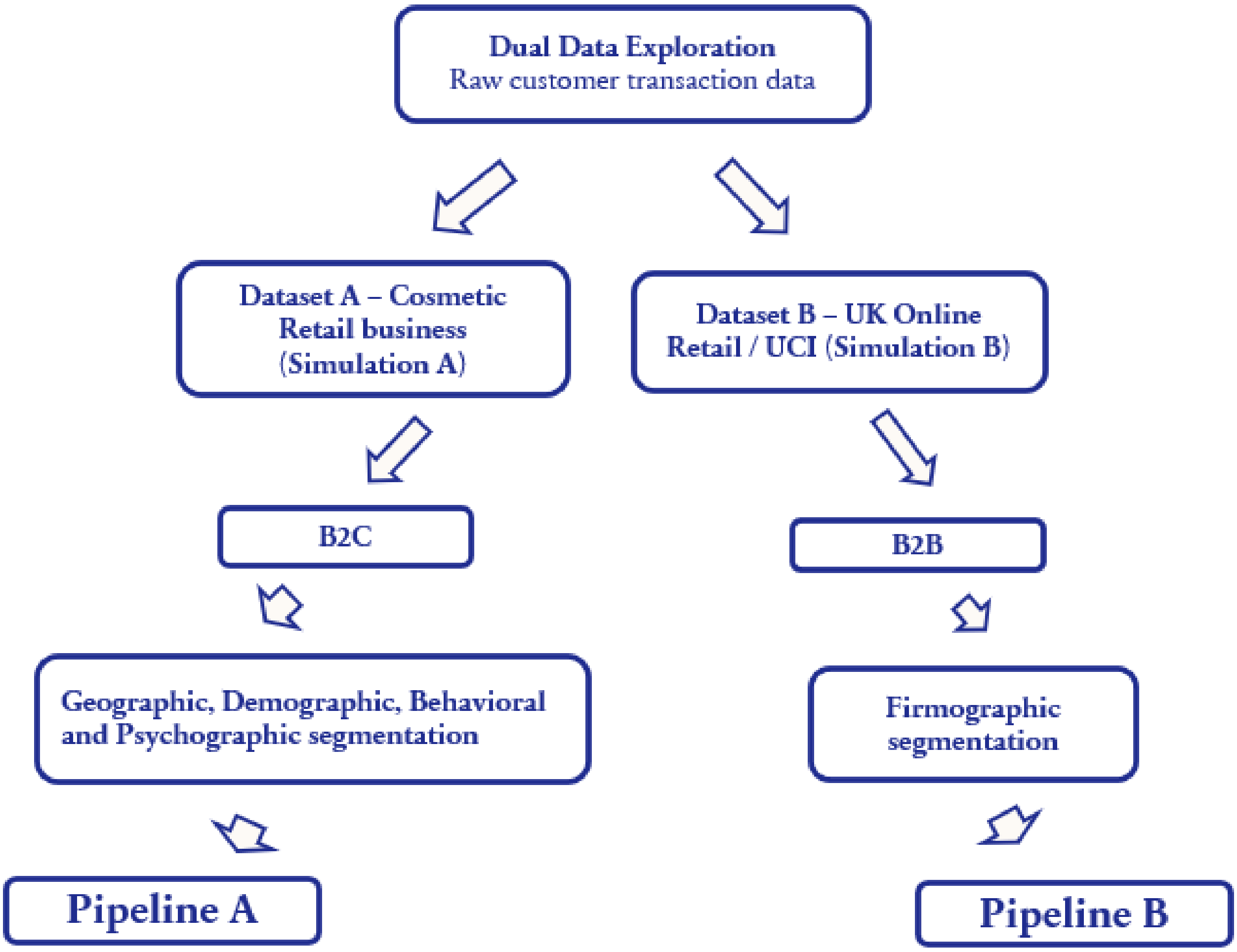

Figure 1: Dual data exploration flowchart showing Dataset A and Dataset B.

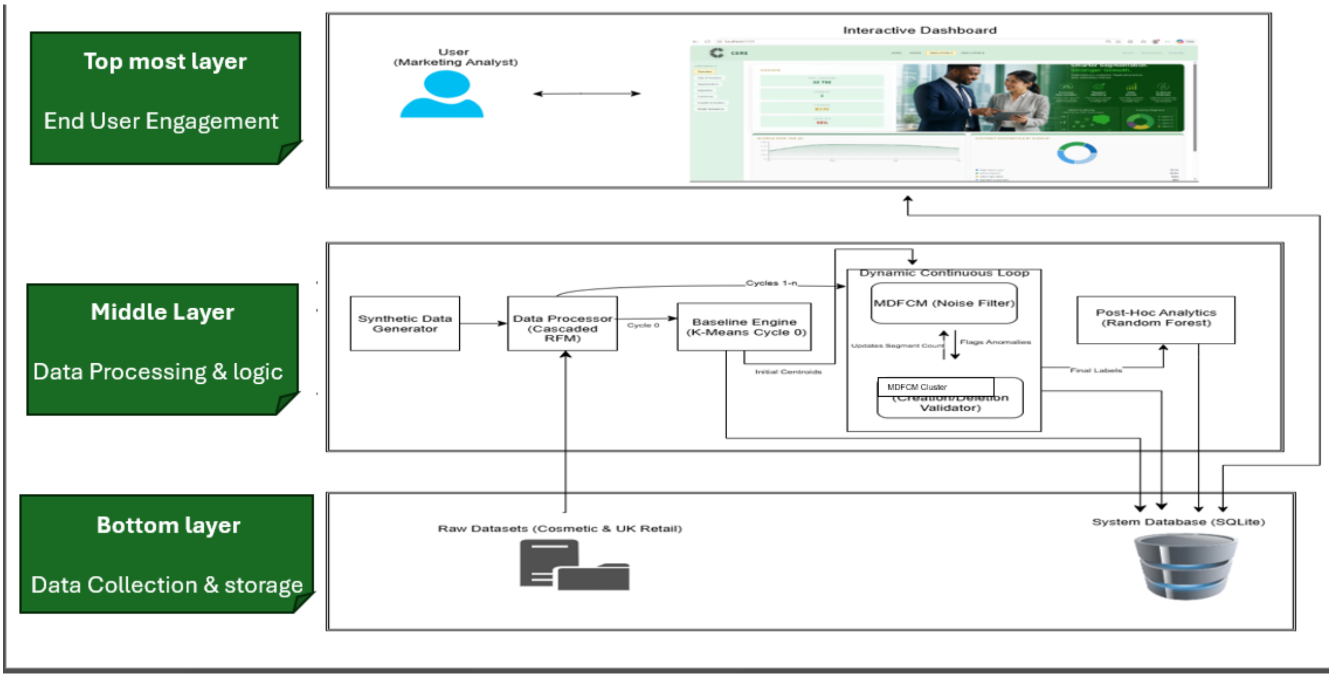

Figure 2: System architecture showing the 3 layers.

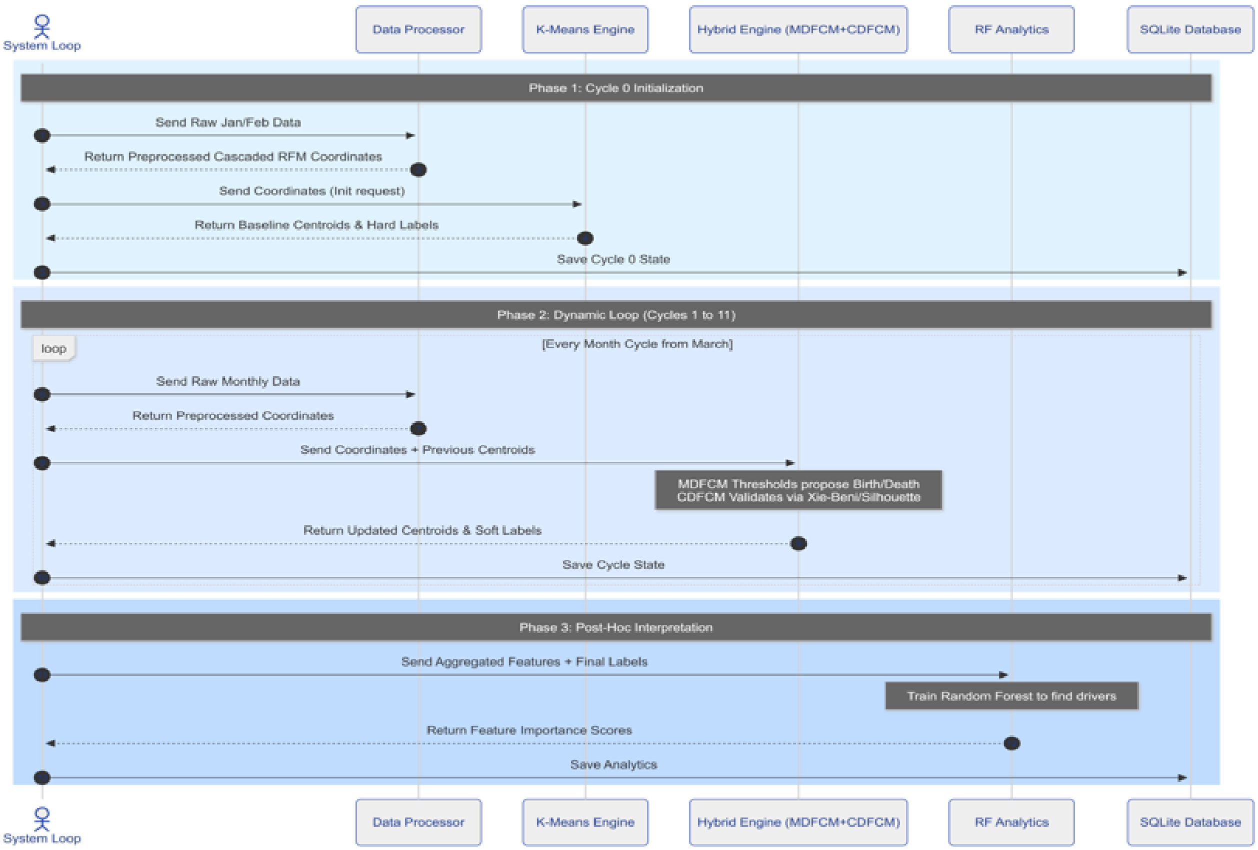

Figure 3: Sequence diagram showing cycle 0 initialisation, monthly dynamic loop updates and post-hoc interpretation.

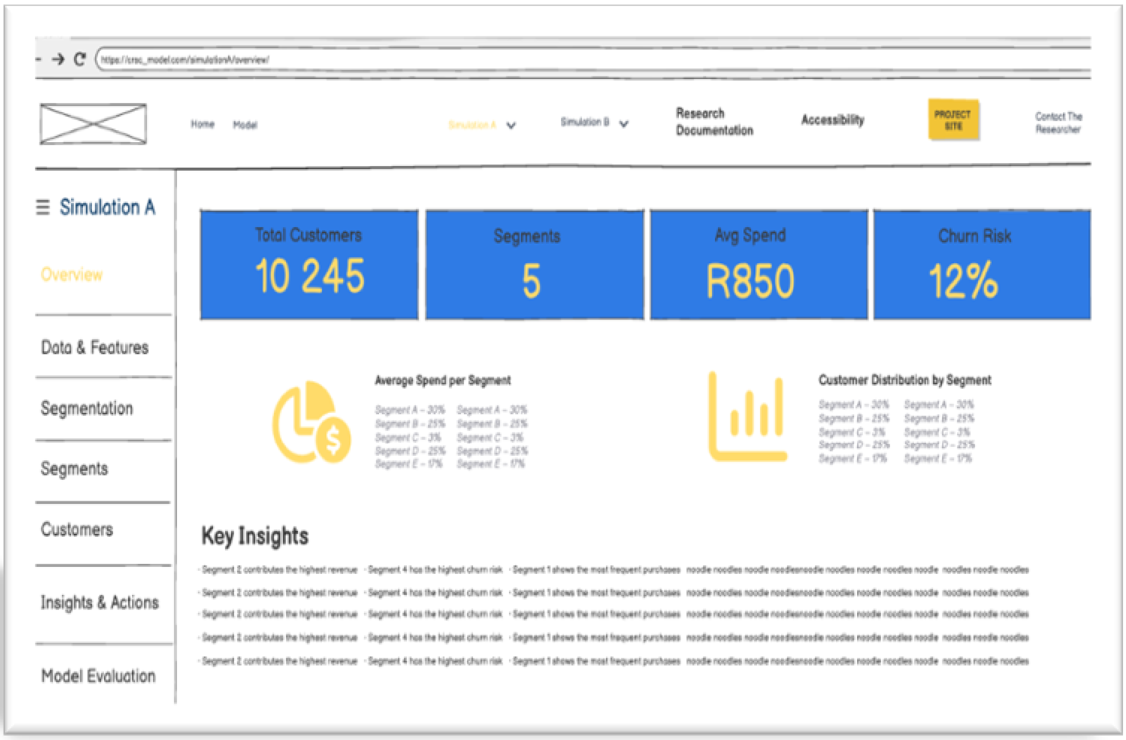

Figure 4: Dashboard wireframe used for the interface design and segment interpretation pages.

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TIMELINE

Phase	**Tasks/ Activities**	**Duration**	**Expected Outcome**
1. Definition (Term1)	Definition of problem statement, aim, objectives, and research questions	Week 1	Approved project scope
2. Literature Review (Term1)	Review customer segmentation concepts, clustering methods, feature engineering, related retail studies and finding the literature gap	Weeks 2–3	Historically used segmentation methods and tools, Literature gaps
3. Data Collection & Understanding (Term1)	Obtain dataset, inspect variables	Week 4	Dataset description and variables understanding
4. Data Cleaning & Preprocessing (Term2)	Handle missing values, duplicates, cancellations/returns, and standardize variables where needed	Weeks 5	Cleaned and analysis-ready dataset
5. Feature synthesis and Engineering (Term2)	Synthesize demographic and geographic features and create segmentation features	Week 8	Final feature set for segmentation
6. Model Development (Term2)	Apply segmentation methods	Weeks 10–12	Customer segments generated
7. Model Evaluation & Interpretation (Term3)	Compare segment quality, interpret cluster characteristics, profile customer groups	Week 13	Final segmentation results and discussion
8. Interface Development (Term2–Term3)	Develop simple interface to display segment outputs, graphs, and customer group comparisons	Week 10–24	Segmentation interface
9. Report Final Edits (Term3)	Compile methodology, results, discussion, conclusions, and references	Weeks 18–20	final report
10. Final Review & Submission (Term3)	Proofread, fix formatting, and submit report/project	Week 21–22	Submitted project
11. Presentation Preparation (Term3)	Design slides, refine visuals, summarize findings, and prepare speaking points	Week 23–26	Final presentation